Paired Audits for Joint Failures Under Changing Incentives

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Abstract

A trained assistant can pass tests that separately change oversight and reward while failing when both change together. We identify the range of joint-only failure rates consistent with paired single-change tests. We give a minimax audit for discovering missed failures and an exact finite confidence bound after zero findings. The design trades off the cost of screening against savings on joint-condition labels. A synthetic learning experiment shows how longer fitting can erase robustness to cue reversal. Balancing the cue only in the joint condition preserves predictive information in the other conditions.

Summary and scope

Question. What do separate oversight and incentive tests leave unknown about harm when both changes occur?

Finding. A uniform audit is minimax for finding failures in the screen-negative pool. A separate stratified calculation bounds total joint harm when both pools are sampled. The fitted experiment and its population counterexample show why joint training coverage alone cannot certify robustness to a new cue distribution.

Evidence and scope. The finite-cohort sampling bounds are proved and checked by exact enumeration. The training evidence comes from synthetic feature models, with a separate exact population optimum. A release evaluator can use the bounds for a fixed cohort and checkpoint. Transfer to new tasks and language models requires further evidence.

1 Two changes a cautious checkpoint can face

A support assistant may respect a veto when it expects an audit or when a shortcut yields no reward. In a new workflow, the audit cue may disappear at the same time that completing a prohibited shortcut raises the score. Separate tests of the two changes establish little about that combination unless the evaluator can justify a structural relation among outcomes. The case-paired design below uses the language of potential outcomes for factorial experiments [1].

Take a fixed finite population of N cases, each including the same task, permissions, checkpoint, and any specified simulation seed across conditions. Let $Y_{ab}(i) \in \{0, 1\}$ indicate an adjudicated harmful action in case i under two switches $a, b \in \{0, 1\}$. For example, switch a removes a visible observer cue and switch b changes which action earns an immediate reward. In separate sandboxed copies of the case, screen both single changes. Set

$$U = \{i : Y_{10}(i) = 1 \text{ or } Y_{01}(i) = 1\}, \quad V = \{1, \ldots, N\} \setminus U, \quad m = |V|.$$

The joint-only set is $D = \{i \in V : Y_{11}(i) = 1\}$. Its size d counts failures that the single-change screens missed, regardless of whether the baseline outcome was harmful. If the intended target requires a clean baseline as well, add baseline failures to U before sampling and state that narrower estimand.

2 What the single-change screens identify

Without a restriction on the relationship between conditions, every integer d from zero through m is compatible with the same complete single-change observations. For each value, choose a d-element subset of V and make the joint condition harmful exactly on that subset, leaving the other three conditions unchanged. This construction applies even if every screened case passes. For example, one policy may always respect a veto while another takes a prohibited action only when both switches are active. Those policies agree on the baseline and on both single-change conditions. More single-change labels cannot distinguish them.

Pairing matters when determining the screen-negative pool. Marginal single-change rates only bound their union. Recording both outcomes per case identifies that union and avoids counting the same screened failure twice. The method does not impose monotonicity, so a single-change failure can also disappear when both switches change.

3 A minimax check for missed failures

Theorem 1. *Suppose $0 \le k \le m$. Among randomized choices of exactly k distinct cases in V, uniform sampling maximizes the worst-case probability of hitting an unknown d-element failure set. The value is*

$$1 - \frac{\binom{m-d}{k}}{\binom{m}{k}},$$

with $\binom{x}{k} = 0$ when $x < k$.

Proof. Fix any sample of size k and average the no-hit event over a uniform d-element failure set. Its probability is $\binom{m-k}{d}/\binom{m}{d}$, equal to the stated no-hit probability by counting disjoint pairs. Averaging over any randomized sampling design gives the same value. Its worst failure set therefore cannot yield a higher detection probability than the stated average. Uniform sampling attains that average for every failure set. □

This theorem assumes the failure set is fixed before the audit draw. A policy that detects the draw or changes between cases requires a different adversary model. The optimum is also a worst-case result across unknown failure locations. A trusted predictor can support a more efficient unequal allocation, provided the inference accounts for that selection.

An exact one-sided confidence procedure follows from the same probability. If the sample is clean, report the largest d with no-hit probability strictly greater than α. If the sample finds a failure, report m as the upper count in this zero-findings procedure. For every fixed D, understating the true count is possible only on a clean sample whose probability is at most α. To bound all joint harm rather than the joint-only subset, add at most $|U|$ to this upper count. A separate probability sample of U could tighten that conservative bound.

4 Bounding all joint harm after zero findings

Divide the cohort into strata of sizes m_i, such as U and V. Within each stratum draw k_i cases uniformly without replacement, independently of the other draws. If d_i cases in stratum i are harmful under the joint condition, the no-find probability is

$$P_0(d_1, \ldots, d_s) = \prod_i \frac{\binom{m_i - d_i}{k_i}}{\binom{m_i}{k_i}}.$$

To bound the total $d = \sum_i d_i$, maximize this expression over all placements of d harms. This can be done greedily. With $r = m_i - d_i > 0$, adding one harm to stratum i multiplies its factor by $(r - k_i)/r$ while the numerator is positive. Use zero when the factor first becomes zero or is already zero.

For each stratum the successive multipliers form a nonincreasing list. The product for an allocation is a product of prefixes of those lists. Taking the largest available multiplier repeatedly selects d globally largest multipliers, with prefix order maintained even under ties. It therefore attains the maximum. The same zero-findings inversion as before, using this maximum, gives a valid upper bound on total harm.

For the 64-case example below, placing all 12 joint labels in the 48 screen-negative cases gives a 95% total-harm upper count of 25 after zero findings. Allocating nine labels there and three to the screen-positive cases lowers it to 12. It matches the upper count under unstratified sampling in this fixture. With arbitrary failure placement and a fixed total label count, the minimax theorem applied to the full population prevents a screen from improving the worst-case detection probability without extra structure. Its value for targeting joint-only failures is a different claim. Exhaustive allocation checks agreed with the greedy calculation on 27,634 probabilities across 2,954 designs.

5 Finite verification and review cost

Consider 64 cases with a screen-negative pool of 48. We compare 12 uniform joint-condition labels in the pool with 12 unstratified labels. The endpoint is detecting a member of the genuine joint-only set. This comparison credits the unstratified design with detection even if additional screening would be needed to classify the discovered failure as joint-only.

Joint-only failures	Targeted detection	Unstratified detection	Labels to match
1	0.2500	0.1875	16
4	0.6973	0.5739	17
8	0.9198	0.8300	17
16	0.9968	0.9788	17

With four failures, the targeted design costs less than the 17-label unstratified design when the full two-condition screen costs less than 5/64 of a joint label per case. If those screens were already part of checkpoint assessment, the incremental cost can be much lower. At level 0.05, 12 zero-findings labels give an upper count of nine joint-only failures with targeted sampling, or 12 with unstratified sampling. Those bounds are for the 64-case cohort. Population transfer needs additional sampling assumptions.

The checker exhausts sample and failure-set pairs on an eight-case population. Across sample sizes and failure counts from one through four, it verifies both directions of the disjoint-pair identity, covering 26,244 pairs. It computes the larger-cohort bounds and match costs in exact rational arithmetic.

6 A fitted model under changing cue reliability

The exact audit describes detection after training. To investigate how training coverage and the feedback target interact, we ran a four-arm synthetic study. The target is binary and the case specification declares whether acting is useful or harmful. A task feature agrees with the target with probability 0.85. An approval cue agrees with probability 0.97 in the baseline and single-change cells, but only 0.03 in the joint cell. These probabilities define the fixture. They are not estimated from agent behavior.

Each arm starts from the same affine logistic classifier trained to predict the cue. The feature map includes the cue multiplied by the joint-condition indicator. That interaction coefficient has zero gradient whenever training omits the joint cell, so its initial value persists during single-change training. All arms use 768 fine-tuning labels per seed and the same optimizer:

- Verified labels in the baseline and two single-change cells.
- Verified labels in all four cells.
- Approval-cue labels in all four cells.
- Verified labels in all four cells, with the cue replaced by an independent coin in the training inputs.

The test fixes the task and true label and crosses the joint cue three ways. It either matches the fine-tuning environment, reverses again, or is independent. Every arm is evaluated on the same cases. Across eight seeds there are 4,800 observations in each joint condition and 14,400 observations across the other three cells. The three joint columns are paired projections, so they should not be pooled as independent test cases.

Training arm	Single cells*	Joint matched	Joint reversed	Joint independent
Single verified	13,962	119	4,681	2,432
Four-cell verified	13,751	4,075	4,075	4,075
Four-cell proxy	13,962	119	4,681	2,432
Four-cell neutralized	12,296	4,075	4,075	4,075

Correct decisions. *Includes the baseline cell.

In the matched joint cell, single-change training and four-cell proxy training each took 2,286 false actions out of 2,347 negative cases and 58 useful actions out of 2,453 positive cases. Four-cell training with either verified target made 335 false actions and 2,063 useful actions. The matched, reversed, and independent scores agree for both verified arms in this run because both made the task-feature decision in the joint cell. Neutralization reduced accuracy in the three cells where the cue remained predictive. The four-cell verified arm preserved more of that accuracy under this feature map and optimization schedule. A different learner or training schedule could learn the conditional cue and lose the observed invariance. The experiment supports a claim about learned decisions in the specified model. Its data do not establish transfer to language models or deployment.

7 Why the fitted invariance need not survive optimization

The data-generating law lets us check what happens at the population optimum. Let $x, z \in \{-1, 1\}$ be the task and cue features and let j indicate the joint cell. The two features are conditionally

independent given the target and cell. Bayes' rule gives the true posterior log odds

$$\log(17/3)x + \log(97/3)(1-2j)z.$$

This function lies in the fitted class. It minimizes population log loss because each conditional log-loss term is minimized by the true posterior probability. Since $\log(97/3) > \log(17/3)$, the resulting joint decision is $-z$, even when the task feature disagrees. Its joint accuracy is 97% in the matched environment, 3% after reversal, and 50% with an independent cue. For neutralized training the posterior log odds is $\log(17/3)x$, giving 85% in all three environments. An independent rational checker enumerates the feature and label cells and reproduces these values.

Thus the finite crossed fit achieved invariance that its own population objective does not preserve. Training to better fit verified labels can still favor a context-specific shortcut. The two tests diagnose different failures. Crossing both switches gives the optimizer evidence about the interaction. The original four-cell distribution and unconstrained log-loss objective have a noninvariant optimum in the stated feature class. Cue neutralization gives one way to change that optimum. A frozen release test should vary the cue again, even after the checkpoint has passed the matched joint condition.

8 Balancing the cue only in the joint condition

Global neutralization discards useful information in the three cells where the cue is stable. Instead, in the joint cell only, replace the training cue with an independent fair coin with probability γ. The resulting joint cue accuracy is

$$r(\gamma) = 0.03(1-\gamma) + 0.5\gamma.$$

The resulting cue remains independent of the task feature conditional on the label. The population score is representable by the same affine model, using the original cue coefficient outside the joint cell and a different one inside it. In the joint cell the score follows the task feature on every input without ties exactly when its log-odds magnitude strictly dominates that of the cue, or $r(\gamma) > 0.15$. Equivalently, $\gamma > 12/47$. At equality some inputs tie. Outside the joint cell the population decision still follows the original cue and has accuracy 0.97.

The task-feature decision is minimax for joint accuracy across the matched and reversed test laws. Their equally weighted mixture makes the cue fair conditional on the label and the task feature. The best accuracy on this mixture is 0.85. For any rule, the smaller of its two accuracies is at most its accuracy on the mixture. Using the task feature attains 0.85 in each law and in the independent-cue test. This argument covers randomized decisions as well. Exact enumeration checks all sixteen deterministic rules on the two binary inputs.

We reused the eight synthetic seeds and their test cases to compare four balancing weights at 60 and 240 epochs. Each joint training row allocates weight $1-\gamma$ to its original cue and $\gamma/2$ to each replacement cue, holding total training weight fixed. The three other cells are unchanged. At 240 epochs the correct counts were as follows.

Joint weight γ	Outside joint	Matched	Reversed	Independent
0	13,962	4,681	119	2,368
1/4	13,962	4,075	4,075	4,075
1/3	13,962	4,075	4,075	4,075
1	13,962	4,075	4,075	4,075

The unbalanced run reproduces the earlier counts at 60 epochs and loses most reversed-cue accuracy with longer training. One-quarter balancing appears robust at 240 epochs but is below the population threshold. Its fitted robustness in this comparison cannot certify the behavior of its objective

at the population optimum. One-third and full balancing satisfy the threshold in the synthetic model. The follow-up uses the original test cases and gives no new external test of generalization.

9 Use during training and release

Screen paired cases while tuning the policy, then freeze a distinct cohort and checkpoint before making a release claim. Draw the targeted joint tests with a seed inaccessible to the acting policy. In an authorization-sensitive workflow, useful outcomes include acting on permitted requests and asking for clarification when authority is ambiguous. Report both along with harms so that a checkpoint that simply refuses everything does not appear useful. A prospective agent study should compare the verified curricula at a common audit cost and hold out entire task families. The synthetic study supplies a control with known cue reliability. An agent study would need external outcome adjudication and validation that both interventions preserve the task and authority.

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References

[1] Tirthankar Dasgupta, Natesh S. Pillai, and Donald B. Rubin. *Causal Inference from* 2^k *Factorial Designs Using the Potential Outcomes Model*. Journal of the Royal Statistical Society, Series B 77(4), 2015. `https://arxiv.org/abs/1211.2481`.